$$\bar{p} = \frac{\sum_i w_i\, p_i}{\sum_i w_i}, \qquad w_i = r_i^{N_i}, \qquad p_i = \begin{cases} (1-r_i)^{s_i} & \text{if got}_i \\ r_i^{t_i}(1-r_i)^{6-t_i} & \text{otherwise}^1 \end{cases}$$

i ranges over the related viruses

got_i true if the ith member has a site at this location

r_i silent mutation rate in the window

s_i the number of mutations that would remove a site

t_i the number of mutations required to add a site

N_i the number of actual silent mutations in the window

1: In the rare cases where more than one site is possible with silent mutations (e.g. sometimes GAGACC and GAGACG) we sum over those, but I haven't written the formula for that for simplicity.